ELAN1 GOVERN · FRONT END · BUILD REPORT

Completeness gaps — built

28 October 2026, night. This report covers every item that the evening check (`09-completeness-check`) marked Missing or Partial, across L1, L2, L3 and Sovereign. The public site is not included.

> **How it was built.** Twelve builders worked in parallel, one per area. Then came a type-check and production build, an independent check of each area against its item list, a fix round for every gap that check found, and a final build. After that, the app was crawled live in the browser at every level.

1. Summary

120	118	2	114	0
ITEMS IN SCOPE	BUILT	PARTIAL	GAPS FOUND BY THE CHECK, THEN FIXED	TYPE OR BUILD ERRORS

The type-check (`tsc -b`) and the production build (`vite build`) both pass. The only build warning is the size of the demo-data chunk, which does not affect the app.

2. Live check

The app was crawled in the browser by following every link, signed in as an admin at each level. Each page was checked for broken links, error pages, "no page at this address" and script errors. The pages outside the doors (status, legal, key fingerprints, maintenance, offline, developer portal, verifier) and every new screen were also opened directly.

LEVEL	SIGNED IN AS	PAGES OPENED	LINKS FOLLOWED OR CHECKED	PROBLEMS
L1 Guarded	Fatima Sheikh (admin), with two-step sign-in through the login form	400	31,227	0
L2 Attested	Neha Bhatia (admin)	400	37,044	0
L3 Proven	Arvind Swamy (admin)	320	27,567	0
Sovereign	Meenakshi Sundaram (admin)	320	30,403	0
Outside the doors and new screens	—	17	—	0

The crawl capped how many pages of the same shape it opened, for example individual receipts. So "pages opened" is a sample of repeated pages plus every distinct screen reached. The live crawl found one start-up error, a module-load-order problem; it is fixed (section 5), and all four crawls ran after the fix.

3. What is still partial

ITEM	WHAT IS MISSING
Pilot tier shown in a demo organisation	The pilot is fully built for organisations created by sign-up. That covers the ₹0 line, the hard stop at 2,000 actions (`TENANT_STOPPED`, with no record written), the 30-day read-only period and the move to the paid plan. Demo Traders shows its pilot history and has *End the pilot now*. The pilot invoice line and the Today tile still appear only in a sign-up organisation. Seeding them into a main demo organisation would freeze writes that every other journey uses.
Seeded demo events carry the seed time	Seeded path-to-production, records and pack events are stamped when the seed runs, not back-dated. Back-dating would make timestamps run backwards in the hash chain. Weaving them into the generated history is a larger change to the chain generator.

4. Defaults to confirm

The documents were silent or disagreed on these points, so the build uses the defaults below. Each can be changed.

QUESTION	DEFAULT BUILT
Emergency stop	One person engages it at any level. Lifting it needs a second, different owner or admin from L2 up, Sovereign included. At L1, one owner or admin lifts it.
Supervision sampling (A3)	A review after the call runs; the call is not held.
Graduation checks	Three checks: 90 days, 200 records and 100 approvals. The precedent score is shown for information only.
Re-proposal after a revoke	30-day cooldown.
Public status page	Sovereign is not shown. Its status lives in the bank's own console.
Public `/verify` samples	Never write to or bill a tenant chain.
Sandbox calls (`sk_sbx_`)	Metered at ₹0; shown on the invoice as a ₹0 line.
Require SAML for everyone	Applies to the organisation's own domains. External auditors keep password plus two-step sign-in. An owner with two-step sign-in is the break-glass exception, and the receipt records it. The demo starts with SAML on but not required.
Same person approving	Flagged at L1, forbidden from L2: the owner of the proposing agent cannot approve its call. Rejecting it is always allowed.
Step-up on approval	₹5,00,000 or more, and every irreversible call.
Reversal window	30 days from the decision, for reversible tools only.
Break-glass access	Two people at every level, L1 included; 30 minutes to 4 hours; expires by itself.
Signing a key at L2 and L3	At L2, turning signing on needs a different owner or admin from the person who bound the key. At L3 and Sovereign, binding needs 3 of 5 key holders.
Key compromise	Records after the declaration are sealed by an interim Elan1 key and labelled "not signed with your organisation's key" until the new key is bound. Notice goes to a named contact within 24 hours.
Bundle size	About 1 MB per bundle (estimated). Larger ranges split, and each part is its own ₹2,500 bundle.
Upgrade rollback	Owner only, before the closing record. The rolled-back period is billed as L1.
Conformance tests, refusal register, MCP tool names	The documents name only some of these. The builders filled in the rest: the conformance tests C1–C14 (the documents name only C14), 27 of the 34 MCP tool names, and the applicant categories. Please review the wording.

5. Fixed after the build

- A start-up error: a module-load-order cycle (`recordedChecks` used before it was initialised) between `mock/db.ts` and `mock/records.ts`. The registries that other modules add to at load time now live on a hoisted holder in `db.ts`, so the cycle can no longer break them.

6. Items by area

1. Sign-in and sessions

ITEM	STATUS	WHERE
Viewer demo sign-ins at L2, L3 and Sovereign	**Built**	src/features/auth/registry.ts (EXTRA: Pooja Malhotra Demo-Viewer-L2-2026, Nandini Hegde Demo-Viewer-L3-2026, Sunita Joshi Demo-Viewer-SOV-2026); listed on /login under each level

Approve-by-link sign-in goes through the same checks as /login	**Built**	New src/features/auth/LinkSignIn.tsx, used by src/features/approvals/ApproveByLinkPage.tsx (/approve/:token)
Sign-out ends the session and writes auth.signed_out	**Built**	src/store/session.ts signOut(opts?) / endSessionRecord; callers TopBar (unchanged call), Sessions.tsx, LoginPage, ApproveByLinkPage, keepSignedIn.ts
Auditor suspension is per organisation	**Built**	src/features/auth/signInPolicy.ts chooseOrg/policyFor; src/store/session.ts switchOrg + syncMembership; LoginPage AccountRow
Forgot password covers portal accounts	**Built**	src/features/auth/registry.ts requestPasswordReset/resetPassword; /forgot and /reset/:token
Portal sign-in is a real portal session	**Built**	registry.ts startPortalSession/currentPortalSession/endPortalSession; new src/features/auth/PortalSession.tsx; src/features/portal/CounterpartyPortalPage.tsx; LoginPage openPortal
SAML 'Require SAML for everyone' forces SSO-only for password sign-in	**Built**	signInPolicy.ts policyFor/samlRequired/breakGlassOwners; RefusalBanner.tsx; Sso.tsx (BreakGlassNote, SignInPreview line); LoginPage 'SAML required' badge
Live-demo traps: admin Unlock and Reset demo data clears sign-in state	**Built**	registry.ts unlockSignIn/signInLock/resetSignInState; Members.tsx (Sign-in locked chip, Unlock sign-in menu item); Sessions.tsx (new Locked sign-ins panel); db.ts onResetDemo hook
Pre-seeded suspended account	**Built**	signInPolicy.ts SEEDED_SUSPENSIONS + ensureAuthSeeds; registry.ts demo account u_nb_deepak (Demo-Suspended-L2-2026)
Backup-approver labels match the escalation seed	**Built**	registry.ts EXTRA journeys for u_nb_imran, u_kh_rafi, u_dc_farhan

An independent check found 2 gap(s) in this area after the first pass; all were fixed in the fix round.

2. Approvals

ITEM	STATUS	WHERE
Decide screen: field-by-field before→after diff with "Show all fields", keyboard reachable	**Built**	/approvals/:id — src/features/approvals/decide.ts (recordDiff), src/features/approvals/DecidePanels.tsx (DiffSection), wired in ApprovalDetail.tsx
Every check's result (G1–G9 and the rule) with its version	**Built**	/approvals/:id — decide.ts checkResults/checkSummary, DecidePanels.tsx ChecksSection
Payload digest of exactly what the approver was shown, displayed and recorded on the decision receipt	**Built**	decide.ts shownPayload/payloadDigest; DigestLine in the action bar; DecisionForm payloadDigest prop → backend decide/decideByLink extra → receipt details.payload_digest; RecordedDigest in the decided block; ApproveByL…
Blast radius with dual control for tenant or cross-boundary scope	**Built**	src/mock/approvalsExt.ts blastRadius; backend.ts needsTwo/twoPersonReason; DecidePanels BlastRadiusSection; header badge in ApprovalDetail; queue rows
Gate summary on queue rows	**Built**	/approvals — ApprovalsPage.tsx Row + DecidePanels CheckSummaryLine
"Payment recorded · proof #N" with Open the record and a real "Reverse this" for 30 days	**Built**	DecidePanels OutcomeBlock + ReverseDialog (in the post-decision banner and the decided block); approvalsExt reverseApproval/reversibility/reversalPlan
Approve and draft a rule	**Built**	ApprovalDetail action bar button; DecisionForm draftRule prop; decide.ts ruleFromApproval/draftRuleFromApproval; approvalsExt recordRuleDraft

Fail-closed "Approvals unavailable" page and outage state, with a demo trigger	**Built**	DecidePanels ApprovalsUnavailable (replaces the inbox and every decide screen on /approvals and /approvals/:id); /approvals/settings 'Approvals service' panel; approvalsExt startOutage/endOutage; backend decideBlocker…
Queue filters in the URL: waiting on me, expiring within an hour, tool, counterparty	**Built**	/approvals?mine=1&soon=1&tool=<tool>&cp=<counterparty> — ApprovalsPage.tsx
Same-actor forbid mode (org setting: flag vs forbid)	**Built**	/approvals/settings 'When an agent's owner is the approver'; approvalsExt sameActorMode/setSameActorMode/sameActorBlocks; backend decisionDetails; ApprovalDetail SameActorBanner; ApproveByLinkPage approve button
Named, time-boxed delegate	**Built**	/approvals/settings Delegates panel; approvalsExt addDelegation/revokeDelegation/delegateFor/delegations; backend mayDecide (delegateFor); db.ts approvalHooks used by decideApproval; ApprovalDetail DelegateBanner; dec…
Step-up for irreversible or high-value approvals	**Built**	/approvals/settings Step-up panel; approvalsExt stepUpPolicy/setStepUpPolicy/stepUpReason/markStepUp/hasStepUp; DecisionForm; backend decisionDetails; BlastRadiusSection note

An independent check found 8 gap(s) in this area after the first pass; all were fixed in the fix round.

3. Work: stop, break-glass, incidents, telemetry

ITEM	STATUS	WHERE
Emergency stop under dual control	**Built**	src/mock/ops.ts (liftNeedsSecond, requestLift, approveLift, declineLift, withdrawLift, pendingLift); src/features/incidents/ResumeFlows.tsx (LiftStopForm + SecondPersonLift); src/features/incidents/IncidentsPage.tsx (…
Break-glass as a visible dual-control write	**Built**	src/mock/ops.ts (requestBreakGlass, approveBreakGlass, declineBreakGlass, withdrawBreakGlass, endBreakGlass, auto-expiry sweep); src/features/work/BreakGlassPanel.tsx; shown in Organisation › Sessions (settings/sectio…
Today due-dates panel (obligations calendar)	**Built**	src/features/today/dueDates.ts; src/features/today/DueDatesPanel.tsx (DueDatesPanel, MaintenanceNotice); src/features/today/TodayPage.tsx
Approver telemetry and rubber-stamp sampling	**Built**	/work/approvers → src/features/work/ApproversPage.tsx; src/mock/ops.ts (approverTelemetry, approverDecisions, drawApproverSample, reviewSampledApproval, noteApprovalOpened); nav item 'Approvers' (owner/admin) in Work …
Supervised 1-in-10 sampled calls as Work items	**Built**	src/features/work/backend.ts workQueues (supervisedSamples); src/features/work/ReviewsPage.tsx (kind supervised_sample, inline Right for A3 / Record a concern via api.reviewSample); src/features/today/TodayPage.tsx (W…
Incident classes (key compromise, isolation breach, declaring by hand) and CERT-In 6-hour clock	**Built**	src/mock/ops.ts (declareIncident, certIn*, saveCertInDraft, submitCertIn, markCertInNotReportable, certInBlocker); src/features/incidents/DeclareDialog.tsx; src/features/incidents/CertInPanel.tsx; IncidentsPage (Decla…
L1 chain or receipt-write failure opens a health incident at L1	**Built**	src/features/work/backend.ts deriveAssurance (chain_integrity and new receipt_write detector run before the L2 gate); src/mock/ops.ts (probeReceiptWrites, startWriteDrill); src/features/alerts/detectors.tsx (chain_int…
System pages wired: /403, /offline, /maintenance	**Built**	src/features/system/RequirePerm.tsx (route gate → /403?perm|roles|what|from) used in App.tsx for /agents/new, /billing, /work/approvers; ForbiddenPage reads level permissions, roles and what; src/features/system/Syste…

An independent check found 12 gap(s) in this area after the first pass; all were fixed in the fix round.

4. Keys and key compromise

ITEM	STATUS	WHERE

Key-compromise flow (L2+): declare, stop signing, keep writing unsigned, show unsigned interval, 24-hour notice to the named contact, rotate to a new key, close	**Built**	src/mock/keys.ts (declareCompromise, sendNotice, bindKey, signingOn, closeCompromise, setContact, unsignedIntervals); src/features/keys/Compromise.tsx (KeyCompromisePanel, DeclareCompromiseDialog, NoticeDialog, BindKe…
Key windows with overlap periods, a revoked status, and refusal of key changes while a chain incident is open	**Built**	src/types.ts (SigningKey.status adds 'revoked' | 'bound'; optional publishedAt, revokedAt, revokedReason, interim); src/mock/keys.ts (keyWindows, keyChangeBlock, assertKeyChangeAllowed, bindKey planned, withdrawBound)…
"Key bound · signing off" state shown during the window between binding and signing	**Built**	src/features/keys/KeyWindows.tsx (banners and chips), src/features/keys/Compromise.tsx (SigningStateBadge on the Keys page tile, compromise step 3), SigningKeys.tsx (bound key line in the active-key panel and the In u…
KMS Sign-call reconciliation (L2): KMS Sign calls vs receipts signed, discrepancies flagged	**Built**	src/mock/keys.ts (reconDays, reconView, runReconciliation, explainDiscrepancy); src/features/keys/SignReconciliation.tsx; /agents/keys?tab=custody
Auditor's evidence request as a recorded write (auditor requests range/bundle; org sees and fulfils; receipts both ways)	**Built**	src/mock/keys.ts (requestEvidence, fulfilRequest, declineRequest, withdrawRequest, collectRequest, settleSeeds); src/features/auditor/EvidenceRequests.tsx (EvidenceRequestsPanel and default page); new route /assurance…
Fingerprint published as a printable page (route) and as a DNS TXT record value with copy	**Built**	public route /fingerprint/:org (org id or slug), src/features/keys/FingerprintPrintPage.tsx; src/features/keys/FingerprintPublish.tsx on /agents/keys?tab=signing
KMS rotation: write the approval receipt only after the rotation succeeds (KeysPage.tsx ~139)	**Built**	src/mock/keys.ts approveKmsRotation and requestKmsRotation; src/features/agentsdoor/KeysPage.tsx; src/features/evidence/KeysCustody.tsx (had the same bug)

An independent check found 11 gap(s) in this area after the first pass; all were fixed in the fix round.

5. Evidence, bundles and the verifier

ITEM	STATUS	WHERE
L1 auditor pack gains a PDF summary, compliance map and README (in-browser PDF, included in pack/ZIP)	**Built**	src/lib/pdf.ts (new minimal PDF 1.4 writer), src/mock/evidence.ts buildAuditorPack/complianceMapOf/summaryPdf/l1Readme; exportAuditorPack in src/mock/db.ts now delegates; ZIP via bundleToZip in src/features/verify/bun…
L1 auditor pack manifest sealed with last hash, file hashes and counts; verifier checks it	**Built**	src/mock/evidence.ts buildAuditorPack (manifest_seal by sealing_kid); src/lib/verify6.ts section 1 (L1_PACK_MEMBERS, l1PackContents)
Framework report as PDF (Coverage/framework pages)	**Built**	src/features/evidence/frameworkPdf.ts; /compliance menu 'PDF (report to file)' (CompliancePage.tsx); /compliance/:framework 'Export PDF' button (FrameworkPage.tsx)
Nightly integrity-run history and continuity report	**Built**	src/mock/evidence.ts (nightlyRuns, integrityRuns, continuityReport, runIntegrityNow, continuityPdf); src/features/evidence/IntegrityRuns.tsx; /evidence?tab=integrity (new 'Integrity runs' tab on Proof); summary panel …
Enforcement mode in every export manifest and on the Assurance door	**Built**	src/mock/evidence.ts enforcementOf/ENFORCEMENT_LABEL; enforcement_mode and enforcement in the L1 manifest (evidence.ts) and the L2 manifest plus VERIFY.txt (platform.ts buildEvidenceBundle); AssuranceHomePage hero 'En…
L2 export guards: oversize split, clean range, residency/retention/consent gate, missing declarations, 'Verify it yourself first'	**Built**	src/features/evidence/ExportGuards.tsx (planParts, firstBreak, GuardsPanel, PreflightResult), src/features/evidence/ExportPackDialog.tsx; api.previewPack (no write)
Sandbox receipts marked as sandbox and shown in Activity/receipt pages	**Built**	Receipt.environment (types.ts); governCall opts in db.ts; ensureSandboxSeed in src/mock/evidence.ts (run once per org from announceOpen); src/features/evidence/EnvironmentChip.tsx; ModeChip in activity/shared.tsx (Act…

Console exports as ZIP (keep JSON option)	**Built**	ExportPackDialog container choice (ZIP default, JSON); bundleToZip/parseFile in features/verify/bundle.ts now handle L1 packs
'Export ready' shows bundle digest; dialog shows ₹2,500 charge for L2+ before generating (L1 free)	**Built**	ExportPackDialog (Charge cell, button label 'Generate and download · ₹2,500' or '· free'; done screen lists bundle digest and file SHA-256 per part; 'Check it here' confirms the digest matches)
'Continuous' never shown with a green tick; 'not checked over this range' state	**Built**	AssuranceHomePage tile 1 (neutral 'Links continuous' plus UncheckedChip), hero line; EvidencePage hero; IntegrityRuns.tsx UncheckedChip
A tampered L1 sample on /verify	**Built**	features/verify/bundle.ts buildSample 'fail-integrity-l1' (edited amount) and 'fail-tail-l1' (cut-off tail); buttons on VerifyPage
/verify samples do not write evidence.exported receipts or bill	**Built**	features/verify/bundle.ts orgBundle/l1Pack now use the pure buildEvidenceBundle (platform.ts) and buildAuditorPack (evidence.ts)
PASS variants 'payload unbound' and 'L1: internal evidence only'	**Built**	VerifyCode in types.ts ('PASS_L1_INTERNAL', 'PASS_PAYLOAD_UNBOUND'); verify6 EXIT_CODES/CODE_LABELS/variants/code precedence; ResultView CODE_META and ExitTable; activity/VerifyResult CODE_MEANING

An independent check found 9 gap(s) in this area after the first pass; all were fixed in the fix round.

6. L1 to L2 upgrade close-out

ITEM	STATUS	WHERE
Separate key-bind step: create/bind the KMS key before the switch, show and compare its fingerprint	**Built**	/assurance/upgrade (stage 'Signing key in AWS KMS' for L1→L2): src/features/assurance/upgrade/KeyBindStage.tsx; mock src/mock/upgrade.ts (createUpgradeKey, compareUpgradeFingerprint, takeBoundKey); platform.applyUpgra…
"Signing on at seq N" named on the confirm screen before the switch	**Built**	/assurance/upgrade confirm step for L1→L2: src/features/assurance/upgrade/SignOnConfirm.tsx; mock preSwitchCheck/signOn/nextSigningSeq; api.switchToL2
First-ten-records fingerprint check after the switch	**Built**	/assurance/upgrade/closeout panel 2: src/features/assurance/CloseoutPage.tsx (FirstTen); mock runFirstTen / firstTenRows
Tenant exports the first bundle and runs a tampered copy that exits 1, on screen	**Built**	/assurance/upgrade/closeout panel 3 (FirstBundle); mock runFirstBundle
Closing record with its seven checks	**Built**	/assurance/upgrade/closeout side panel 5 (Closing); mock checksFor / writeClosingRecord
Signed-island rollback (roll back a failed upgrade, recorded)	**Built**	/assurance/upgrade/closeout side panel 'Roll back a failed upgrade' plus a 'Signed islands' list; mock rollbackUpgrade; CloseoutCard on /assurance/levels
Customer notices: 14-day notice to a named contact, boundary-day confirmation, auditor note, composed, scheduled and recorded	**Built**	New wizard stage 'Customer notices' (L1→L2 only, before 'Second person') and /assurance/upgrade/closeout panel 4: src/features/assurance/upgrade/NoticesPanel.tsx; mock scheduleNotice/sendNoticeNow/cancelNotice/dispatc…

An independent check found 4 gap(s) in this area after the first pass; all were fixed in the fix round.

7. Records and data protection

ITEM	STATUS	WHERE
Consent ledger (DPDP): consents per data principal and purpose, withdrawals, linked to the G1 consent gate, on a screen under Records	**Built**	Route /records/consent (nav: Records › Consent ledger). UI: src/features/records/ConsentPage.tsx. Mock: src/mock/records.ts (consentView, recordConsent, withdrawConsent, g1Check, purposesFor), exposed as api.records.*

Erasure plan preview (fields, systems, exports affected) before opening a request; refusal recorded when statutory retention conflicts (GST/MCA)	**Built**	/records/erasure (src/features/erasure/ErasurePage.tsx, new src/features/erasure/plan.tsx). The record-page erasure dialog in src/features/records/actions.tsx is updated too. Mock: previewErasurePlan, openErasureWithP…
Withdraw a declaration version (recorded; new calls stop using it)	**Built**	New route /records/declarations (src/features/records/DeclarationsPage.tsx), dialog in src/features/records/declarations.tsx, a 'Withdraw this version' menu item in src/features/rulebooks/VersionsTab.tsx. Mock: withdr…
Chain gap as a sealed gap record, visible on Chain and Proof	**Built**	src/features/records/gaps.tsx (ChainGapsPanel and the 'Seal a gap record' dialog), inserted on /activity/chain (ChainPage.tsx) and on the /evidence Overview tab (EvidencePage.tsx). Mock: sealGap, gapsView in src/mock/…
RBI model-documentation export (from AI inventory)	**Built**	'RBI model documentation' button on /inventory (InventoryPage.tsx) and /inventory/:id (AiSystemPage.tsx). ModelDocDialog in new src/features/inventory/model.tsx. Mock: exportModelDoc, modelDocHistory in src/mock/recor…
Model name and version on agents and AI systems (fields, forms, display, seeded)	**Built**	types.ts: Agent.modelName/modelVersion and AiSystem.modelName/modelVersion (optional). Form fields in RegisterSystemDrawer.tsx (register and edit; the edit Patch in inventory/ops.ts) and in the AgentNewPage.tsx step 1…
Hosting profile 2: dedicated residency domain option (L2+), shown on Organisation/Deployment and in exports	**Built**	New src/features/hosting/HostingProfilePanel.tsx, inserted on /agents/deployment (DeploymentPage.tsx) and /agents/organisation?tab=residency (OrganisationPage.tsx). Mock: hostingView, requestDedicated, activateDedicat…

An independent check found 10 gap(s) in this area after the first pass; all were fixed in the fix round.

8. Regulatory packs and profiles

ITEM	STATUS	WHERE
Pack admission flow and portfolio register: real gate evaluation (not hard-coded Met), candidate and rejected packs listed	**Built**	/records/packs/portfolio → src/features/packs/PortfolioPage.tsx; gate scoring evaluateGates/GateFacts/PACK_FACTS in src/lib/packs.ts; register + propose/admit/reject in src/mock/packsFactory.ts (portfolio, proposeCand…
Pack detail laid out as the factory's 11 blocks	**Built**	/records/packs/:id → src/features/packs/PackPage.tsx (rewritten), blocks.tsx, ProfileBlock.tsx, TestsBlock.tsx, PilotBlock.tsx; template data PACK_BLOCKS/PACK_TEMPLATE/PACK_WRITE_TYPES in src/lib/packs.ts
T1–T8 test results per pack; pilot block; 90-day pilot tracker (day 0–85, necessity statement, day-76 self-export, verdict capture), starting at L1	**Built**	TestsBlock.tsx and PilotBlock.tsx (blocks 10 and 11); evaluateTests/runPackTests/recordDrill/recordReading and startPilot/recordNecessity/recordSelfExport/recordAirGap/recordVerdict/recordGd/stopPilot in src/mock/pack…

Separate author and certifier roles; certification flow; profile statuses including withdrawn; P1–P12 results	**Built**	/assurance/profiles → src/features/profiles/LifecyclePanel.tsx (new, shown in ProfilesPage); conformance/submitProfile/recordSecondReader/runProfileFixtures/issueProfile/withdrawProfile/profileLifecycle in src/mock/pa…
Two-way reconciliation against the target system's own audit trail	**Built**	src/features/packs/ReconcilePanel.tsx (on the pack page; also the pilot's day-78 step); reconcile/explainException in src/mock/packsFactory.ts
not_measurable and out_of_scope dispositions	**Built**	ProfileRule.disposition/reason/wouldChangeIf/adr and ProfileRuleResult.state additions in src/types.ts; runProfileRules and dispositionProblems in src/lib/profiles.ts; chips in ProfilesPage.tsx and ProfileBlock.tsx
Verifier-profile entitlement enforced; seed profile orders for L2+ demo orgs with packs installed	**Built**	entitlement/guardRun/guardClaim/orderProfile/cancelProfileOrder/ensureSeed in src/mock/packsFactory.ts; enforced in api.runProfile, api.exportPack, api.previewPack (src/api/index.ts); ordering UI in LifecyclePanel, Pr…
Uninstall/cancel a pack (recorded; billing stops)	**Built**	'Remove' on PackPage.tsx (FormDialog); uninstallPack/removedPacks in src/mock/packsFactory.ts
Profile run from a pack page: role check, and L1 preview runs not recorded	**Built**	src/features/packs/ProfileBlock.tsx (block 8); server-side guardRun in api.runProfile

An independent check found 18 gap(s) in this area after the first pass; all were fixed in the fix round.

9. L3: autonomy and counterparties

ITEM	STATUS	WHERE
Selective-disclosure bundle export over a range (one file)	**Built**	/assurance/disclosure?view=bundles (src/features/disclosure/BundlesView.tsx); public portal /counterparty/<org>.rb.<id> (src/features/portal/BundlePortal.tsx); src/mock/l3disclosure.ts; src/lib/disclosure.ts (Disclosu…
Saved disclosure profiles (auditor, lender presets)	**Built**	/assurance/disclosure?view=profiles (src/features/disclosure/ProfilesView.tsx); offered in FieldPicker (src/features/counterparties/shared.tsx) in the package composer, bundle composer and Issue-a-receipt dialog; src/…
Replay detection (nonce check, RECEIPT_REPLAYED)	**Built**	src/lib/crossorg.ts (replayOf, replayStep, REPLAY_CODE); recorded check in src/mock/platform.ts checkReceived; browser check and 'Try a resent copy' in CounterpartiesPage ReceivedDrawer; StandingChip in counterparties…
Dual-control receipt accept (second person; API role check)	**Built**	CounterpartiesPage ReceivedDrawer 'Accept it' section and ReceivedActionDialog; src/mock/l3crossorg.ts acceptReceived/refuseAcceptance; api.acceptReceived / refuseReceived (role checked in src/api/l3extras.ts and in t…
Volume and reversal warning before the first issue to a new audience	**Built**	FirstIssueWarning and useFirstIssue in src/features/counterparties/shared.tsx; shown in the Issue-a-receipt dialog, the disclosure package composer and the bundle composer; api.firstIssueInfo / issueToCounterpartyChec…

Revoke and dispute states for issued cross-org receipts	**Built**	CounterpartiesPage IssuedDrawer (Revoke / Record a dispute / Resolve) and ReceivedDrawer (Raise a dispute / Close our side); DisputeBlock and StandingChip; portal dispute and standing on /counterparty/:token (src/feat…
Recorded check of a received receipt uses the published/pinned key, never the key inside the receipt	**Built**	src/mock/platform.ts checkReceived
Autonomy: automatic revocation on breach (autonomy_breach incident revokes)	**Built**	src/mock/l3autonomy.ts (enforceAutonomyBreaches, reportBreach, autonomyBreachesView); BreachesPanel and Report dialog in src/features/agentsdoor/AutonomyBreaches.tsx, shown on /agents/autonomy and on the agent's Auton…
Autonomy: 30-day cooldown after a revoke before re-proposal	**Built**	src/lib/autonomy.ts (COOLDOWN_DAYS, cooldownFor); enforced in src/mock/platform.ts graduationCandidates (whyNot, plus cooldownUntil and cooldownAutomatic on the candidate), so proposeGraduation refuses; shown in Auton…
Autonomy: precedent score shown as information beside the three graduation numbers	**Built**	src/lib/autonomy.ts precedentScore (precedent-score/1); RecordChecks in src/features/agents/AgentAutonomy.tsx, given a record prop everywhere it is used (agent tab, Propose and Decide dialogs, AutonomyPage agent block…

An independent check found 9 gap(s) in this area after the first pass; all were fixed in the fix round.

10. Billing and the meter

ITEM	STATUS	WHERE
Unreconciled period holds the invoice (status "held")	**Built**	/billing (Invoices table, invoice drawer, Reconciliation panel 'By period'); src/mock/billing.ts (reconRunsOf, latestRuns, heldPeriods, rerunReconciliation); src/features/billing/chainMeter.ts (ChainInvoiceStatus 'hel…
Dispute settlement outcome: upheld (credit note), rejected, partly upheld	**Built**	/billing invoice drawer (held line, 'Record elan1's decision' SettleForm); src/mock/billing.ts settleDispute and settlementsOf; chainMeter.ts creditNotesFor (credit_note documents in the invoice list)
Discount line with an end date	**Built**	/billing Discounts panel (src/features/billing/TermsSections.tsx DiscountsPanel and DiscountForm); invoice lines of kind 'discount' in chainMeter.ts; src/mock/billing.ts recordDiscount and endDiscount
Pilot tier: 90 days free, 2,000-action hard stop (TENANT_STOPPED)	**Partial**	src/mock/billing.ts (PILOT, pilotOf, startPilot, convertPilot, pilotGate); guard hook at the top of governCall in src/mock/db.ts; /billing PilotPanel; pilot invoice lines; pilot rule in the price-book rules list
"Modelled list price" label with target dates; "+4% a year after March 2029"	**Built**	/billing header chip, Current plan panel (ListPriceNote), plan comparison row (PlanSections.tsx), rules list, invoice PDF note; src/mock/billing.ts LIST_PRICE and listPriceFactor
Overage discount cap	**Built**	src/mock/billing.ts checkDiscount and DISCOUNT_CAP; DiscountForm validation; Discounts panel footer and rules list
Cost per governed action published beside the refusal share	**Built**	/billing 'Published every month' panel (TermsSections.tsx PublishedMonthly); src/mock/billing.ts costPerAction
Today "Meter" tile agrees with /billing	**Built**	src/features/today/TodayPage.tsx (levelExtras and LevelTiles meter tile); src/features/billing/summary.ts loadMeter and meterFacts, shared with BillingPage
Pack and profile lines pro-rated by install/order day like the floor	**Built**	src/features/billing/chainMeter.ts buildInvoices; src/mock/db.ts deriveBilling (small edit)
Add-ons evidence count counts only L2+ bundles at ₹2,500	**Built**	src/features/billing/BillingPage.tsx AddOnsPanel and data loader
Sandbox (sk_sbx_) calls metered at 0	**Built**	chainMeter.ts (ChainEvent.sandbox, units 0, left out of months, days, FY, totals and refusalCost; ₹0 'Sandbox calls' invoice line); src/mock/platform.ts meterView (quantity 0, chain recount skips sandbox); pilot count…

An independent check found 10 gap(s) in this area after the first pass; all were fixed in the fix round, except as noted in section 3.

11. Developer portal and path to production

ITEM	STATUS	WHERE
Conformance suite C1–C14 and conformance levels CL-1…CL-D	**Built**	/developers/conformance (src/features/developers/sections/Production.tsx: Conformance, RunResult); console /agents/production?tab=conformance (src/features/developers/console/ProductionPage.tsx); logic in src/mock/dev…
Offline-verifier release page (SHA256SUMS, signature, test vectors, tampered bundle)	**Built**	/developers/verifier-release (sections/Production.tsx: VerifierRelease; src/features/developers/release.ts)
Refusal register (POLICY., GATE., PLATFORM. families) with a page per code	**Built**	/developers/refusal-codes (register with search and filters) and /developers/refusals/:code (one page per code, by wire spelling or REF number); sections/Register.tsx; data in src/mock/devpathData.ts (REFUSAL_CODES, F…
Glossary	**Built**	/developers/glossary (sections/Register.tsx: Glossary; GLOSSARY in src/mock/devpathData.ts)
Action Receipt standard: licences, governance, errata, certification	**Built**	/developers/standard (sections/Production.tsx: Standard)
Built-in MCP tool catalogue (read.*, stage.write, act.* with scopes)	**Built**	/developers/mcp-tools (sections/Register.tsx: McpTools; MCP_TOOLS, MCP_SCOPES in src/mock/devpathData.ts)
Deprecation and Sunset headers in the versioning policy and in TryIt	**Built**	/developers/changelog#deprecation (sections/More.tsx); TryIt POST /v1/check key selector (src/features/developers/TryIt.tsx); DEPRECATIONS and deprecationHeaders in src/mock/devpathData.ts
Support tiers	**Built**	/developers/support (sections/Register.tsx: Support; SUPPORT_TIERS, SEVERITIES in src/mock/devpathData.ts)
Changelog in date order	**Built**	src/features/developers/sections/More.tsx (CHANGES sorted newest first, stable within a day)
TryIt 'open this rulebook' link uses the signed-in org's real rulebook slug	**Built**	src/features/developers/TryIt.tsx (useRulebookLink)
Self-serve sandbox tenant (metered at 0)	**Built**	Console /agents/production?tab=sandbox (ProductionPage.tsx: SandboxApply, SandboxActive); src/mock/devpath.ts (applySandbox, verifySandbox, sendFixture, sandboxUsage)
Production-key request with human review (request → review queue → approve/decline, 2-day target) instead of direct gk_ issue	**Built**	Console /agents/production?tab=keys and ?tab=review (ProductionPage.tsx: KeysTab, ReviewTab, ReviewRequest; console/RequestForm.tsx); src/mock/devpath.ts (requestKey, requestProblems, reviewRequest, collectKey, withdr…
Promotion review advisory → enforced (≥30 days, bypass proof)	**Built**	Console /agents/production?tab=promotion and ?tab=review (ProductionPage.tsx: PromotionTab, PromotionDialog, ReviewPromotion); src/mock/devpath.ts (advisoryEvidence, requestPromotion, reviewPromotion)
Onboarding refusal sentences for the 8 applicant categories	**Built**	src/features/onboarding/refusals.ts (APPLICANT_REFUSALS); company step of OnboardingPage.tsx; sandbox application (ProductionPage.tsx); /developers/path-to-production#who-we-refuse
Developer path to production docs page	**Built**	/developers/path-to-production (sections/Production.tsx: PathToProduction); nav entries in src/features/developers/nav.ts
Demo data seeded in each demo org	**Built**	src/mock/devpath.ts (seed, run lazily on first read of devpathState)

An independent check found 14 gap(s) in this area after the first pass; all were fixed in the fix round, except as noted in section 3.

12. Status, notices, Sovereign operations

ITEM	STATUS	WHERE

/status: SLO cards, error-budget policy and burn, meter and receipt-writer components	**Built**	/status — src/features/status/StatusPage.tsx, src/features/status/parts.tsx (shared row/legend components), src/mock/statusSlo.ts
Elan1-to-tenant notices: named contact, 30-minute incident updates, post-incident note, daily chain-head receipt (hosted orgs)	**Built**	/agents/notices (nav item 'Notices from Elan1', Agents door) — src/features/status/NoticesPage.tsx, src/mock/tenantNotices.ts
Legal pages: privacy notice, cookie policy, sub-processor list (linked from sign-up and footer)	**Built**	/legal/privacy, /legal/cookies, /legal/sub-processors (/legal redirects) — src/features/legal/LegalPage.tsx; links in verify/PublicChrome footer, developers/PublicChrome footer, auth/AuthLayout footer, SignupPage term…
Mobile PWA manifest, icons and installability	**Built**	public/manifest.webmanifest, public/sw.js, public/icons/{icon-192,icon-512,icon-maskable-512,apple-touch-icon}.png, index.html
Sovereign: Channels, You and Webhooks respect deny-all egress	**Built**	src/features/work/ChannelsPage.tsx, src/features/you/YouPage.tsx, src/features/settings/sections/Webhooks.tsx, src/features/settings/sections/Notifications.tsx (embedded in Channels), src/features/deployment/EgressBlo…
Sovereign: Deccan removed from Elan1's public hosted /status	**Built**	src/mock/platform.ts statusView (skip sovereign chain components and approvals-outage names), src/features/status/StatusPage.tsx (custody, anchors and chain rows only for hosted orgs, plus a note); local status in /ag…
Sovereign: egress log (blocked attempts with destination, time, component)	**Built**	/agents/deployment?tab=egress — src/features/deployment/EgressLog.tsx, src/mock/sovereign.ts
Sovereign: carry the meter to Elan1 with no egress — signed usage statement exported on media, recorded	**Built**	/agents/deployment?tab=meter — src/features/deployment/UsageStatements.tsx, src/mock/sovereign.ts (meterFor, exportUsageStatement, recordHandover, checkStatement)
Sovereign: rollback of an applied update bundle (recorded)	**Built**	/agents/deployment (Updates panel) — src/features/deployment/DeploymentPage.tsx RollbackDialog, src/mock/sovereign.ts rollbackUpdate; types.ts UpdateBundle optional rolledBackAt/rolledBackBy/rollbackSeq/rollbackReason
Sovereign figures match the docs: FIPS 140-3 and 8-year retention	**Built**	src/mock/platform.ts deploymentView (sovereign branch)

An independent check found 7 gap(s) in this area after the first pass; all were fixed in the fix round.

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7. All choices the builders made

The builders' own notes, kept for review. Section 4 lists the ones that most need a decision.

Sign-in and sessions

- SAML 'Require SAML for everyone' applies to the organisation's own addresses: its org domain or any domain listed for OIDC. External auditors such as Kavita at desaico.in are not in the org's identity provider and keep password plus two-step sign-in.
- Break-glass exception, from the Sso.tsx help text 'Keep a break-glass owner with a password and two-step sign-in': an active owner with two-step sign-in may use a password while SAML is required. The two-step code is then mandatory and auth.signed_in records break_glass=true. Dual control for break-glass was left to the security area.
- The SAML seed was changed to not required at L2, L3 and Sovereign, so all demo password accounts keep working until an owner switches it on. Older saved state that has enforced=true with no identity.updated receipt saying so is quietly set back to not required, because that flag never had any effect before.
- The seeded suspension is applied when Northbank's records first open, not baked into db.ts. It writes a real member.suspended receipt by Neha Bhatia dated at that time, and it also covers browsers with saved state from up to 12 hours earlier.
- Portal receipts use actor 'system', because portal people are not console users; their name and employer are in the receipt details. Portal sessions last 12 hours.

- The sign-in policy lives in a new shared module, src/features/auth/signInPolicy.ts, used by /login, /approve/:token and registry.checkPassword. checkPassword now checks the password itself instead of calling api.signIn, so a correct password for a suspended account no longer counts as a failed attempt.
- The OIDC SSO-only check still runs before the password is checked, with no refusal receipt, as before. Refusals after a correct password write auth.sign_in_refused.

Approvals

- New state lives on the org record as an optional approvalsExt field (src/mock/approvalsExt.ts), defaulted on read so older saved localStorage state still loads; nothing was added to types.ts.
- Same-actor default: 'flag' at L1 (Demo Traders) and 'forbid' at L2 and above (Northbank, Kaveri, Deccan). Owners and admins can change it.
- 'Same actor' means the approver owns the agent that proposed the call. Forbid blocks approving only; rejecting is still allowed.
- Step-up default: amounts at or above ₹5,00,000, plus every irreversible call. In the seeded queue the ₹6,12,000 payment and the emails need step-up, while the ₹2,40,000 payment only asks when the sign-in is over 10 minutes old.
- Blast radius classes: external transfers or EXT accounts are 'cross-boundary'; production, access-changing or bulk tools are 'whole organisation' (tenant); money, external sends or a counterparty make 'one account'; everything else is 'single record'. Tenant and cross-boundary trigger two-person approval at every level, so the seeded production deploy now needs two engineers.
- The approvals outage is per organisation and only a demonstration: owners and admins trigger it from /approvals/settings, and /status reflects it. Ending it moves every waiting approval's expiry, and unused links', forward by the downtime, which is how 'nothing expires while this is down' is implemented.
- Reversal window is 30 days from the decision. Reversible: payments, bank-detail changes, vendor creation, deploys, merges and generic writes. Irreversible: external transfers, emails and deletes. Reads have nothing to reverse. Checks are called 'checks' on screen (never 'gate'), with G1–G9 ids and declaration versions.
- Demo seeding runs once per org, deferred with setTimeout so nothing is written during a render: one live delegation (the procurement approver's role to the EU finance lead's role, 4 days) and one already-reversed approved payment decided 2–20 days ago, each with its own receipt.

Work: stop, break-glass, incidents, telemetry

- Emergency stop: engaging is one person at every level; lifting needs a second, different owner/admin from L2 up including Sovereign; at L1 one owner/admin lifts. The second person types RESUME and re-enters the password if their sign-in is older than 10 minutes. api.liftStop now refuses at L2+.
- Break-glass needs two people at every level, L1 included. Access is only for an active owner or admin. The approver must be neither the requester nor the person who gets access. The time box is 30 minutes, 1, 2 or 4 hours, and access expires by itself (breakglass.expired, written by the system). Its effect: password sign-in is allowed while the organisation is set to SSO only or SAML required.
- Security incidents use kind 'health' with new subkinds key_compromise / isolation_breach / security_other, labelled 'Security · …', so the Incident['kind'] union in types.ts is unchanged.
- CERT-In: key compromise and isolation breach always start the 6-hour clock from the notice time; other declared classes start it if ticked. Only an owner may record 'not reportable', and never for those two classes. A qualifying incident cannot be resolved until the submission or the 'not reportable' record exists.
- 'Decided without opening' means the approver did not open /approvals/:id before deciding. Opening is tracked from now on by SystemWatch. For approvals decided before this change, opening is filled in deterministically from a hash of the approval id, with Slack and WhatsApp decisions mostly not opened.
- The approver telemetry page is for owners and admins only (route gated to /403). Sampling weights fast decisions and decisions made without opening; nobody samples or reviews their own decisions.
- Pack pilot end = the pack's own pilotEndsAt if another area adds one, otherwise installedAt + 30 days.
- Maintenance windows are platform-wide, not per organisation, and stored in localStorage key eg:state:ops:maintenance. Owners can start the next planned window early from Today, labelled as a demonstration, and end it from /maintenance.
- Receipt-write detection in the demo: a write probe to browser storage during 'Run health checks', plus a two-minute write-failure drill. The chain_integrity and receipt_write detectors now run from L1.
- Seeded history: lifted on read through opsExt on OrgDB, which is optional and defaulted on read. Seeded items carry no receipt numbers because receipts cannot be backdated.

Keys and key compromise

- The unsigned interval is implemented as an interim Elan1 Govern chain key (SigningKey.interim) that seals records hash-chain-only, so existing verifiers keep passing. The UI labels these records 'not signed with your organisation's key' rather than leaving signatures empty, which would have broken verifiers in several other areas.
- The compromised key gets status 'revoked' with retiredAt set to the declaration time, so existing key-window checks reject any later use. The key.rotated receipt that stops signing is signed by the interim key, not the compromised one.
- Chain incident means an active db.incidents entry with subkind 'chain_break' or 'receipt_write'. An open key compromise also blocks ordinary key changes, but not its own bind and signing-on steps. Declaring a compromise is never blocked.
- Each refused key change is recorded as a key.change_refused receipt before the error is thrown.
- At L2, turning signing on needs a different owner or admin from whoever bound the key (dual control, like the existing KMS rotation). At L3 and Sovereign, binding needs 3 of 5 key holders.
- The 24-hour notice goes to a per-org named contact for key notices (seeded as the org's admin who maps to Rahul Menon; editable, and the change is recorded), with an optional copy to the auditor of record. The emails go to the demo outbox via pushMail.
- Planned successors use a 30-day overlap by default (the IM's 'successor 30 days ahead'), selectable as 7, 14 or 30 days. Seeded L2+ keys get publishedAt up to 30 days before signing; L1 keys get no overlap.
- Sign-call reconciliation uses IST days over a 14-day window: receipts plus checkpoints signed at L2+, interim-key records excluded. It is shown at L2 (CloudTrail) and with HSM audit-log wording at L3 and Sovereign; L1 gets an upgrade path.
- DNS TXT format: name _elan1-govern.<domain>, value "v=EG1; kid=<kid>; alg=ed25519; fp=<16 hex>". The printable page is public at /fingerprint/<slug or org id> and shows public keys only.
- Evidence requests: only auditors (member role 'auditor') request; owners and admins fulfil or decline. Fulfilling runs the real export. A bundle collected after a page reload is regenerated for the same range and recorded as a new export.
- All new state lives on OrgDB as an optional keysExt, with every field defaulted on read, so saved localStorage state from before this change still loads. Seeds are written without emitting events. The seeded open evidence request's receipt is written only on the first asynchronous read, never during a render.

Evidence, bundles and the verifier

- The bundle size limit is an estimate of 1,024 KB per bundle (3.1 KB per sealed record, 1.0 KB per L1 record), so the split shows on Northbank's whole chain. Each part is its own bundle, its own evidence.exported receipt and its own ₹2,500 line (billing counts receipts by format, unchanged).
- The L1 manifest is sealed with the org's current Ed25519 key (manifest_seal, sealing_kid). L1 copy says 'sealed', never 'signed'. Packs made before this are accepted with a note, not failed.
- Nightly runs are worked out from the chain (02:00 IST; hashes, links, checkpoints since the previous run) and are not stored. Runs on demand also check every signature or seal in the verifier worker, and are stored in OrgDB.evidence.integrityRuns plus an evidence.integrity_run receipt.
- The unanchored tail is treated as a range problem that is not blocking, with a 'clean range' up to the last anchored seq. Chain breaks and failing pre-flight results block the export until the range is changed.
- Missing declarations (agent without an AI system, no approved assessment, unsigned rulebook version) block the export until acknowledged. The residency domain comes from api.custody. A transfer basis is needed only when the recipient is outside it, and a purpose only when personal_data records are in range.
- Sandbox demo data: 4 non-'ask' calls through governCall with environment 'sandbox' for the first active agent that has an sk_sbx_ key, once per org. Billing does not yet exclude sandbox decisions (chainMeter is another area); flag this if sandbox calls should not be billed.
- The L1 compliance map counts the receipts in the pack against base controls EG-01 to EG-13 (small predicate table in mock/evidence.ts) and maps them to db.frameworks that are switched on. It does not import features/compliance into the mock layer.
- PASS_L1_INTERNAL applies only to auditor-pack/1 files, or manifests with assurance_levels that are all L1. A bare receipt list does not get it.
- The manifest field holding the residency, retention and consent answers is named export_checks (not 'gate', which is banned in customer copy).

L1 to L2 upgrade close-out

- All close-out state comes from receipts in each org's own chain, grouped into an 'episode' that starts at the newest level.upgrade_started for L1→L2. Only the bound key's private seed and the full notice text and address are kept outside the

chain, in an optional OrgDB field (upgradeCloseout) that defaults on read.

- New code lives in new files: src/mock/upgrade.ts, src/mock/upgradeSeed.ts, src/api/upgradeCloseout.ts, src/features/assurance/CloseoutPage.tsx and src/features/assurance/upgrade/{shared,KeyBindStage,NoticesPanel,SignOnConfirm,CloseoutCard}.tsx.
- The new flow applies only to L1→L2 runs. L1→L3 and Sovereign keep the old KMS stage and confirm screen. If an upgrade happens outside the wizard (for example during onboarding), a new key is created at the switch as before.
- The switch writes signing_on (seq N) plus nine level.signing_canary records, so the first-ten check has ten signed records at once instead of waiting for traffic.
- After the switch the key can no longer be replaced; using a different key means cancelling and starting again. Before the switch, a new key may replace an uncompared one, and the replacement is recorded.
- Switching earlier than the date in the 14-day notice is allowed only with a recorded tick that the named contact agreed. The seeded Northbank close-out uses this path, because its chain window starts only five days before the boundary.
- Rollback removes the L2 period from levelHistory rather than adding an L2→L1 period. This keeps boundaries, anchoring and seals consistent everywhere else, and means the rolled-back period is billed as L1. The island is recorded in the level.upgrade_rolled_back receipt and listed on the close-out and Assurance levels pages.
- Rollback is owner only and allowed only before the closing record while the org is still on L2. Running the checks, the first bundle, the closing record and the notices is open to owners and admins.
- Demo seeding: Northbank (L2) has a complete, closed close-out seeded into its chain; the smoke test showed all seven checks passing live. Demo Traders (L1) runs the whole flow live. Kaveri and Deccan show an explanatory empty state.
- Notices go to email addresses; receipts store only the mail domain and a SHA-256 digest of the body. Boundary-day and auditor bodies use placeholders that are filled in when the notice is sent.
- The L1 claim rule is kept: copy about records before the boundary says 'sealed in a hash chain'. 'Signed' is used only for the L2 key and for the island of L2-era records.
- Verification: I bundled a Node smoke test with esbuild in the scratchpad (no dev server). On Demo Traders it confirmed the bound key signs, signing starts at the named seq, the first ten pass, the bundle exits 0 and the tampered copy exits 1, notices are sent, the closing record is refused while anchoring waits, and rollback works. On Northbank all seven checks pass. The screens have not been checked in a browser.

Records and data protection

- A 'declaration version' is taken to be a rulebook version, because Seal F07 declaration_version is 'the rulebook version in force'. Withdrawal is an optional `withdrawn` field rather than a new RulebookStatus value, so existing draft and publish logic (status !== 'published') is not affected.
- Withdrawing the only version in force is refused (publish a replacement or pause the agent first) rather than leaving the agent with no rulebook. Withdrawal needs an owner or admin and a reason; it is not dual-control.
- Retention mapping: write types containing bill, invoice, payment, journal, ledger, credit_note, voucher, gstin or vendor fall under GST (6 years) and MCA (8 years); transfer, bank_details or ifsc fall under PMLA (5 years). Read verbs (list_, read_, get_, search_) hold nothing. A request is refused only when every matching record is held; otherwise it opens for the erasable records and lists the held ones.
- Erasure requests from the console now go through api.records.openErasure, which writes the same erasure.requested action with the plan fields added. platform.createErasure is left untouched for any other caller.
- Most new state lives under db.platform.recordsLedger, created and defaulted on read so older saved state still loads. Model fields go directly on the shared Agent and AiSystem objects as optional fields; withdrawal markers on RulebookVersion.withdrawn. Organisations created by sign-up are not seeded.
- Demo seeding runs once per org on the first read of any records feature and appends seed receipts at the chain head with the current time; past timestamps are never written. The seed receipts are consent.imported, chain.gap_sealed, erasure.refused, rulebook.version_withdrawn (Northbank, Kaveri) and hosting.dedicated_requested/activated (Northbank).
- Hosting profile number: Sovereign is 4, L3 is 3 (with an optional dedicated domain), an L2 org with an active dedicated domain is 2, otherwise 1. Only an owner can request, cut over or return; provisioning completes as soon as it is requested, and the owner confirms the cut-over.
- The consent ledger and G1 are not wired into governCall in db.ts, to keep shared-file edits small. G1 is shown through the tester, the per-consent reliance counts (hashes of the identifiers in decision receipts) and the collect1 pack link.
- The hosting profile is added to the L2+ evidence bundle manifest only, because profile 2 starts at L2. The L1 auditor pack is unchanged.
- Who can do what: consent writes, gap records and model edits need an owner or admin. The RBI export is open to every role except approver.

Regulatory packs and profiles

- The pack-factory state for each organisation lives on db.platform.packFactory, cast the same way features/records/ext.ts does it. Every field is defaulted on read, so platform.ts was not edited and old saved state still loads.
- Profile orders reuse the existing billing.addon_ordered/addon_cancelled receipts (addon:'profile'), so the meter and invoices bill them unchanged. Seeded orders are written lazily at the head of the chain with today's time, not back-dated, so a seeded organisation's profile line starts this month.
- Ordering a profile is owner-only (billing.manage), to match BillingPage. Running needs evidence.export (owner, admin or auditor). Issuing needs owner, admin or auditor and is never done by the author. Withdrawing, uninstalling and drills need owner or admin. The G-D answer is owner-only.
- Screen copy says 'Issued' and 'issuer' for certified and certifier, because 'certified' is a banned word. Reference refusal codes use a CHECK.* prefix, not GATE.*, because 'Gate' is banned in customer copy.
- Each profile now has its own limits statement, so P10 (unique, over 200 characters) is real. Profile digests changed as a result; bundles exported earlier note that they claim a different version.
- The fixture bundles for T5–T7 and P3–P6 are built from the organisation's last 120 receipts. The non-conforming bundle is a sound chain with its checkpoints left out, claiming in-mca-audit-trail, so C04 fails. The incomplete bundle drops declarations.json. These only run at L2+.
- Reconciliation targets follow the demo tools: ledger1 and invoice1 against Zoho Books, collect1 against the Gmail sent log. The ERPNext named in the spec is not in the demo data.
- The portfolio register is shown per organisation. close1 and gxp1 come from the memorandum's 'earlier portfolio of ten'. The 'general AI governance product' rejection is listed as governance1.
- Not changed (other areas) — follow-ups for their owners: (1) CompliancePage.tsx:214 and DecisionTrace.tsx:215 show a 'provisional' gate in red as not met; they should treat state !== 'not_met' as passing. (2) BillingPage's own Order/Cancel calls db.logAdmin directly and skips the withdrawn and L2 checks; it could call api.orderProfile/api.cancelProfileOrder instead. (3) chainMeter builds pack lines only from installed packs, so after an uninstall the recomputed past months lose the pack line (the legacy db.ts invoices keep issued months). (4) ExportPackDialog and ExportRangeDialog list profiles that are not ordered; the server now refuses those claims with a clear message, and api.profiles rows now carry entitlement for filtering.

L3: autonomy and counterparties

- Precedent score formula (precedent-score/1): round(100 × approvals / (approvals + rejections)) − 5 per reversal, clamped to 0–100. Shown as information only and never used for eligibility.
- Cooldown is 30 days after any revocation of a graduated grant, whether manual or automatic, and blocks re-proposal through graduationCandidates.whyNot.
- A breach revokes only graduated (A3) grants of the named agent (and tool, if named). The A2 starting level is untouched. Resolving the breach incident does not restore the grant.
- Who can report a breach: owners, admins and approvers. Declared-by-hand incidents (ops.ts DeclaredClass) do not include an autonomy_breach class, so the Report a breach button and any incident with subkind 'autonomy_breach' are the triggers.
- Replay rule: the first-arrived copy of an issuer+nonce is kept and later copies fail with RECEIPT_REPLAYED. The customer copy shows the bare code with no 'Gate' prefix.
- Dual control: owners and admins only, two different people, and only for verified, unreplayed, unrevoked, undisputed copies. A boundSeq is not required, so seeded pending acceptances can be completed.
- First-issue warning is enforced only through the new *Checked API methods and createDisclosureBundle. The old api.issueToCounterparty and api.createDisclosure are left unenforced so other areas' screens (activity ShareDialogs, records actions) keep working; they should switch to the checked methods and FirstIssueWarning.
- Revoking an issued receipt also withdraws its linked disclosure package, so the portal link stops.
- Disputes: receivers in this console raise them and they are mirrored to the issuer. The issuer can record a dispute from an outside counterparty. A signed-in portal user can raise one from the portal. Disputes are resolved on the issuer side (upheld revokes the receipt); the receiver can close its own side.
- Bundle limits: at most 200 receipts per bundle; expiry 1–365 days (7/30/90 in the UI, or from a profile). Owners, admins and auditors can download; owners and admins can sign and withdraw.
- All new state lives in an optional platform.l3x (src/mock/l3state.ts) with seed-once flags, so saved browser state from before this change loads and gets the seeds on first read.

Billing and the meter

- Every commercial term is stored as an admin receipt in the org's chain (billing.* actions); there is no new OrgDB field, so saved state from older builds still loads. Seeding runs once per org from announceOpen, and its receipts carry details.seeded=true.
- The invoice status 'held' and the credit-note status 'credit' are local to the billing area (ChainInvoiceStatus in chainMeter.ts). The shared Invoice['status'] in types.ts is unchanged.
- Dispute settlements are recorded in the console by an owner, as elan1's decision with a 'Decided by' field defaulting to 'elan1 billing desk', because there is no elan1-staff console.
- A discount applies only to governed actions beyond the allowance (the last N committed events of each segment that fall inside the discount's dates). The floor, refusals, packs, profiles, seats and evidence bundles are never discounted.
- The pilot counts production governed actions (decision receipts, one each, sandbox excluded). The hard stop writes one billing.pilot_stopped receipt. Later refusals with POLICY.TENANT_STOPPED return an unrecorded receipt object (seq 0, rule_id POLICY.TENANT_STOPPED) and are counted in localStorage under eg:data:billing-pilot-stopped:<org>.
- The pilot applies to sign-up (fresh) orgs at L1. None of the four demo orgs was made a pilot, so their established invoices stay as they are.
- Cost per governed action is elan1's modelled cost to serve (runtime cost plus 15%) from the financial model: ₹0.092 in Oct 2026 falling straight-line to ₹0.052 from Apr 2030. It is not the customer's effective price.
- List-price target dates come from the price book: L1 GA Q2 2027, L2 Apr 2028, L3 Oct 2029, Sovereign shown with L3 (Oct 2029). The +4% a year applies each April from April 2029.
- Demo seeds: Northbank has a 15% lead discount (1 Jul 2026 to 30 Jun 2027) and its previous month held by an unmatched reconciliation. Kaveri has a partly upheld floor dispute with a ₹10,000 credit note. Deccan has a 25% founder discount for 12 months. All four orgs have a matched nightly run for the open month.
- I removed a 'you paid per action' column from the monthly panel. With the small chain sample (floor ÷ a few actions) it read as hundreds of rupees per action, which was misleading.

Developer portal and path to production

- Refusal register counts: the documents disagree, so I used 43 core codes (14 POLICY, 19 GATE, 10 PLATFORM), 28 active from L1 and 15 from L2, with the 6 L3 receipt codes added as REF-44 to REF-49. Five GATE codes, three PLATFORM codes and two POLICY codes (AGENT_PAUSED, ARGUMENTS_INVALID) are my additions, made to reach those counts and to map the built system ids.
- No customer prose calls a check a 'gate': the GATE family is labelled 'Pre-write checks' and wire spellings stay as identifiers.
- MCP tool names: only read.records, read.evidence, read.declaration, read.meter, stage.write, act.write and act.escalate come from the documents. The other 27 names were made up to fill the 18/10/6 split.
- C1–C14 are not enumerated in the documents beyond C14, so I defined them. C6, C7, C9 and C11 come from client statements on a REST client (true by design for MCP and SDK). The other tests are worked out from chain data.
- The CL level reached depends on the org's assurance level (CL-1 at L1, CL-3 at L2, CL-4 and CL-D at L3) and on C10 passing.
- The 8 applicant categories and their refusal sentences are my wording; the documents do not list them.
- Reviewer: hosted orgs are reviewed at an 'Elan1 review desk' tab, recorded under the actor 'Elan1 production review (Meera Iyer)' and labelled as shown for the demo. On Sovereign the review is the customer's own: an owner or admin who is not the requester.
- Sandbox tenant: kept as state on the org (devpath field, read with a default) with sandbox receipts on the org chain marked environment: sandbox, not a separate org or chain.
- Deprecation demo: the eg_live_/eg_test_ key prefixes (decision 6h), deprecated 2026-10-28, sunset 2027-10-28.
- Support tiers, verifier release history and errata dates are my content.
- Demo data is seeded lazily on the first read of path-to-production state (opening the page, an agent's Issue key dialog, or the public conformance sample), not when the org opens, to avoid editing db.ts. The seeded receipts are timestamped at seed time.
- No edits to src/types.ts, src/api/index.ts, src/mock/db.ts or src/mock/platform.ts: devpath state lives on OrgDB through a cast, and the screens call devpathApi from src/mock/devpath.ts directly.

Status, notices, Sovereign operations

- No api/index.ts edits: the new calls go through feature-local wrappers (src/features/deployment/sovereignApi.ts and a small noticesApi in NoticesPage.tsx) over new mock modules, following the existing deploymentApi.ts pattern.
- New per-org state is kept in optional fields on the org DB (OrgDB.sovereign, OrgDB.tenantNotices) by cast, without changing the OrgDB interface. Each is seeded on first read, so state saved before this change still loads and persists with saveState.

- The internal allow-list is separate from the internet allow-list, which stays empty (deny all). It accepts only .internal/.local/.lan names, the org's own domain, or RFC1918 addresses, and is seeded for Deccan with siem/smtp-relay/itsm.deccanbank.internal.
- Background egress attempts are generated deterministically per 7-minute slot from the Sovereign start date (stable across reloads). Only attempts made from the console (webhook to an outside URL) are receipts (egress.blocked); firewall background noise is a log, not receipts. The Deployment 'Blocked in 24 hours' figure now comes from this log instead of platform's random number.
- Email and phone links stay available in Sovereign (internal mail relay, console over the bank's network); only Slack, WhatsApp and outside webhook URLs are blocked.
- Rollback moves the bundle running now back to 'staged' (still verified, can be re-applied) and makes the previous applied bundle current, without adding a new status value to the UpdateBundle union.
- Chain-head statements are signed with a separate Elan1 notice key (ntc_2026_10). At L1 the copy describes the tenant chain as 'sealed in a hash chain'.
- SLO measured values other than 'writes without a receipt' (live check) and availability (from the error-budget ledger) are fixed, labelled as targets 'measured over the last 30 days'. The anchor SLO is shown as Missed because of the eMudhra incident, which keeps the page consistent with its own incident log.
- Notices are derived from the same two incident windows /status uses: eMudhra yesterday 09:14 IST for 158 min (shown to L2+ orgs only), and the WhatsApp delay 6 days ago (shown only to orgs with WhatsApp-verified members). The post-incident note is shown as sent 20 h after resolution and as 'due by' before then.
- The PWA service worker caches only the start page and shell assets (network-first navigations) so records are never served stale. It is registered from an inline script in index.html in secure contexts only.
- The 'Notices from Elan1' nav item is visible to all roles; only the named contact, owners and admins can acknowledge, and only owners and admins can change the contact.